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1. Write a C++ program to find the difference between two matrix.

2. How will you pass two dimensional array to a function explain with example.

Structures

Learning Objectives

After the completion of this chapter, the student will be able to

- Understand the purpose of user defined data types
- Able to construct C++ programs using structures
- Execute and debug programs with structure data type

12.7 Structures Introduction

Structure is a user-defined which has the combination of data items with different data types. This allows to group of variables of mixed data types together into a single unit.

12.7.1 Purpose of Structures

In any situation when more than one variable is required to represent objects of uniform data-types, array can be used. If the elements are of different data types,then array cannot support. If more than one variable is used, they can be stored in memory but not in adjacent locations. It increases the time consumption while searching. The structure provides a facility to store different data types as a part of the same logical element in one memory chunk adjacent to each other.

12.7.2 Declaring and defining structures

Structure is declared using the keyword 'struct'. The syntax of creating a structure is given below.

```
struct structure_name {
        type member_name1;
        type member_name2;
} reference_name;
```

Objects declared along with structure definition are called global objects

An optional field reference_name can be used to declare objects of the structure type directly.